

PRECAST PANEL W10-45: MASTER TABULAR COMPARISON & ALGORITHM SPECIFICATION

Independent Ground-Truth Tables (BOM, BBS Rebar, Mirrored Opening Chains, 7-Tier Hierarchy) & Engineering Guidelines for Correct Drawing Generation Algorithms

EXECUTIVE BRIEF FOR ENGINEERING & ALGORITHM TEAMS

Project: PAISLEY TOWER | Piece Mark: W10-45 | Drawing: P22-132

Lead Quality Assurance: Senior Precast Engineering QA & Computer Vision Data Auditor

Reference Files Audited:

- Raw Machine PDF: W10-45_combined_macro.pdf (Sheets 1 to 3, Scale 1:50, 1296 x 864 pt)
 - Engineered Ground Truth PDF: P22-132.W10-45.Rev 2.pdf (Sheets 1 to 4, Scale 1:75, 1224 x 792 pt Tabloid)
- Key Geometric Finding: Panel W10-45 is the exact OPPOSITE-HAND (MIRRORED) structural mate of W10-175. Opening layout is reversed: 2,115 mm (left pier) | 975 mm (window) | 4,928 mm (center pier) | 975 mm (window) | 373 mm (right pier). Overall length is identically locked at 9,366 mm, solid thickness at 254 mm, and panel weight at 24,827 LBS. Audit Status: 100% PASS — CERTIFIED AS PRODUCTION GROUND TRUTH FOR MACHINE LEARNING TRAINING.

PART I: Exhaustive Tabular Comparisons (Ground Truth vs. Macro)

Table 1: Hardware Bill of Materials (BOM) Table Reconciliation (Item-by-Item)

Reconciliation of the complete Hardware BOM table located in the top-right metadata zone (Sheet 1). Constrained strictly to the SCHEDULE_TABLE spatial zone [X >= 1000, 320 <= Y <= 550], the true table displacement is uniformly dx = -35.15 pt, dy = -36.00 pt across all 12 items (σ = 0.000 pt), confirming rigid block translation without phantom drift.

Mark	Category	Description	Macro Qty	Rev 2 Qty	Macro Pos (X, Y)	Rev 2 Pos (X, Y)	True Shift (dx, dy)	Status
P-205	Lifter	12000LB FORGED ANCHOR	4	4	(1058.23, 358.99)	(1023.08, 322.99)	(-35.15, -36.00)	100% Preserved
GT75-686	Grout Tube	75Ø x 686 GROUT TUBE	2	2	(1058.23, 372.49)	(1023.08, 336.49)	(-35.15, -36.00)	100% Preserved
GT75-1219	Grout Tube	75Ø x 1219 GROUT TUBE	8	8	(1058.23, 385.99)	(1023.08, 349.99)	(-35.15, -36.00)	100% Preserved
GT76-254	Grout Tube	76Øx254mm G.TUBE	6	6	(1058.23, 399.49)	(1023.08, 363.49)	(-35.15, -36.00)	100% Preserved
GT100-1219	Grout Tube	100Øx1219 GROUT TUBE	1	1	(1058.23, 412.99)	(1023.08, 376.99)	(-35.15, -36.00)	100% Preserved
P-300	Bracing	19Ø INSERT (NC)	5	5	(1058.23, 426.49)	(1023.08, 390.49)	(-35.15, -36.00)	100% Preserved
P-602	Plate	13PL. W/STUD	1	1	(1058.23, 439.99)	(1023.08, 403.99)	(-35.15, -36.00)	100% Preserved
SP15-457	Splicer	15Mx457 WILLIAMS SPL	2	2	(1058.23, 453.49)	(1023.08, 417.49)	(-35.15, -36.00)	100% Preserved
SP25-1067	Splicer	SP25Mx1067 WILLIAMS	8	8	(1058.23, 467.49)	(1023.08, 431.49)	(-35.15, -36.00)	100% Preserved

Mark	Category	Description	Macro Qty	Rev 2 Qty	Macro Pos (X, Y)	Rev 2 Pos (X, Y)	True Shift (dx, dy)	Status
		S			466.99)	430.99)	36.00)	Preserved
SP30-1219	Splicer	30Mx1219 WILLIAMS SPL	1	1	(1058.23, 480.49)	(1023.08, 444.49)	(-35.15, -36.00)	100% Preserved
CNDT	Conduit	19MM CONDUIT	1	1	(1058.23, 493.99)	(1023.08, 457.99)	(-35.15, -36.00)	100% Preserved
EB-S-PL	Box Plate	Single Plastic Electr	1	1	(1058.23, 507.49)	(1023.08, 471.49)	(-35.15, -36.00)	100% Preserved

Table 2: 7-Tier Dimension Hierarchy Verification & Applied Engineering Logic (Sheet 1)

Complete verification of horizontal dimension chains positioned above the panel face (+Y outward normal direction) following the Separation of Trades Rule for precast production on Panel W10-45.

Tier Level & Name	Target Components	Macro State (W10-45)	Ground Truth State (Rev 2)	Defect Mode in Macro	Applied Engineering Logic
Tier 1 (Innermost)	Lifters (P-205)	Dedicated string missing; lifting anchors mixed with secondary embeds	927 1989 2748 1947 1755 mm Labeled: P-205 LIFTERS (Sum = 9,366 mm)	Hierarchy Inversion	Rigging Priority: Crane riggers and stripping crews require immediate, dedicated pick-point access.
Tier 2	Splicers (Couplers)	Couplers bundled with grout tubes and opening edges without trade distinction	220 1664 1436 1336 2254 292 292 1580 288 mm Labeled: SP15, SP25, SP30	Layer Pollution	Coupler Isolation: Structural shear rebar continuity sleeves isolated for cage tying teams.
Tier 3	Bracing Inserts (P-300)	Placed at erratic vertical level; obscured under generic label 'HARDWARE'	1950 5066 2350 mm Labeled: BRACING INSERTS (P-300) (Sum = 9,366 mm)	Functional Misclassification	Functional Grouping: 5,066 mm brace stance provides optimal torsional stability during erection.
Tier 4	Embed Hardware (Grout Tubes)	Severe bounding-box clash: GT76-254 and GT75-1219 callout tags overlap directly	4100 3294 1971 mm (Sum = 9,365 ≈ 9,366 mm) with staggered leaders at right margin	Visual Collision	Leader Staggering: Alternating vertical elbow offsets (ΔY stagger) eliminate visual clashes.
Tier 5	Secondary Hardware (Plates/Conduits)	Generated unverified phantom mark 'BCMBD-1-K' not in project Bill of Materials	Clean sub-lines for EB-S-PL and CNDT; phantom mark BCMBD-1-K completely purged	Data Corruption	BOM Reconciliation: Systematic dictionary cross-check suppresses unreferenced orphan marks.
Tier 6	Profile Openings (Cutouts)	Fragmented into 2 disconnected lines: Line 1: 2115 975 4928; Line 2: 975 373	Unified continuous string: 2115 975 4928 975 373 mm (Sum = 9,366 mm)	String Fragmentation	1D Span Merge: Mirrored opposite-hand profile unified into a continuous pier/window chain.
Tier 7 (Outermost)	Profile Overall (Envelope)	Intermediate details, leader elbows, and secondary tags cross	Single clean 9366 PROFILE OVERALL string locked at outermost	Boundary Breach	Outer Envelope Invariance: Preserves external formwork boundary constraint without

Tier Level & Name	Target Components	Macro State (W10-45)	Ground Truth State (Rev 2)	Defect Mode in Macro	Applied Engineering Logic
		boundary line	envelope		interference.

Table 3: Bar Bending Schedule (BBS) Table Reconciliation (Complete 10 Marks)

Complete geometric, dimensional, and positional verification of all 10 bent rebar marks migrating from Macro Sheet 2/3 to Rev 2 Sheet 4 for Panel W10-45. Notice that mark RB25M 0141 has leg B=2090 mm, and RB10M 0050 has Qty 13.

Mark	Size/Grd	Type	Len(mm)	Bending Dimensions (mm) [A, B, C, D, E, F, G, H]	PinØ	Qty	Macro Pos	Rev 2 Pos	Row Shift (dx, dy)
RB10M 0040	10M 400W	T1	3790	A=75, B=160, C=1675, D=160, E=1675, F=75	60	10	(77.46, 110.00)	(42.31, 71.96)	(-35.15, -38.04)
RB10M 0046	10M 400W	17	760	A=330, B=160, C=330	60	19	(77.46, 123.50)	(42.31, 81.17)	(-35.15, -42.33)
RB10M 0049	10M 400W	T1	890	A=85, B=190, C=195, D=190, E=195, F=85	40	12	(77.46, 137.00)	(42.31, 90.39)	(-35.15, -46.61)
RB10M 0050	10M 400W	Unk	4240	U-Bar Edge: 630, 335, 1230, 100, 655, 665, 160	60	13	(77.46, 150.50)	(42.31, 99.60)	(-35.15, -50.90)
RB10M 0051	10M 400W	TS5_1	750	A=255, B=255, C=255	60	8	(77.46, 164.00)	(42.31, 108.81)	(-35.15, -55.19)
RB10M 0052	10M 400W	TS67_2	1090	A=100, B=160, C=280, D=160, E=280, F=160, G=100	40	13	(77.46, 177.50)	(42.31, 118.02)	(-35.15, -59.48)
RB10M 0125	10M 400W	14	860	A=200, B=140, C=210, D=140, E=200, H=100, H2=100, O=405	60	4	(77.46, 191.00)	(42.31, 127.23)	(-35.15, -63.77)
RB10M 0135	10M 400W	Unk	1690	Legs: 515, 100, 760, 195 (Angle: 45°)	60	22	(77.46, 204.50)	(42.31, 136.44)	(-35.15, -68.06)
RB15M 0068	15M 400W	Unk	1140	Legs: 520, 165 (Hook: 90°)	60	2	(77.46, 218.00)	(42.31, 145.66)	(-35.15, -72.34)
RB25M 0141	25M 400W	17	2380	A=210, B=2090, C=210	150	1	(77.46, 231.50)	(42.31, 154.87)	(-35.15, -76.63)

Table 4: Straight Rebar Schedule Reconciliation (All 18 Line Items)

Verification confirms that all 18 straight longitudinal and transverse rebar types from Macro Sheet 2 & 3 are 100% accounted for in Rev 2 Sheet 4 for W10-45 without missing a single bar or altering any cut length.

Bar Size	Type Code	Cut Length (mm)	Macro Qty	Rev 2 Qty	Steel Grade	Drawing Placement in Rev 2	Preservation
10M	STR	170	2	2	400W	Sheet 3 Elevation Mat / Sheet 2 Sections	100% Preserved
10M	STR	600	12	12	400W	Sheet 3 Elevation Mat / Sheet 2 Sections	100% Preserved
10M	STR	2020	12	12	400W	Sheet 3 Elevation Mat / Sheet 2 Sections	100%

Bar Size	Type Code	Cut Length (mm)	Macro Qty	Rev 2 Qty	Steel Grade	Drawing Placement in Rev 2	Preservation
							Preserved
10M	STR	2620	6	6	400W	Sheet 3 Elevation Mat / Sheet 2 Sections	100% Preserved
10M	STR	2870	6	6	400W	Sheet 3 Elevation Mat / Sheet 2 Sections	100% Preserved
10M	STR	8790	2	2	400W	Sheet 3 Elevation Mat / Sheet 2 Sections	100% Preserved
10M	STR	9270	3	3	400W	Sheet 3 Elevation Mat / Sheet 2 Sections	100% Preserved
15M	STR	170	2	2	400W	Sheet 3 Elevation Mat / Sheet 2 Sections	100% Preserved
15M	STR	230	1	1	400W	Sheet 3 Elevation Mat / Sheet 2 Sections	100% Preserved
15M	STR	2370	2	2	400W	Sheet 3 Elevation Mat / Sheet 2 Sections	100% Preserved
15M	STR	2560	4	4	400W	Sheet 3 Elevation Mat / Sheet 2 Sections	100% Preserved
15M	STR	2620	14	14	400W	Sheet 3 Elevation Mat / Sheet 2 Sections	100% Preserved
15M	STR	2880	14	14	400W	Sheet 3 Elevation Mat / Sheet 2 Sections	100% Preserved
15M	STR	3200	2	2	400W	Sheet 3 Elevation Mat / Sheet 2 Sections	100% Preserved
15M	STR	4750	4	4	400W	Sheet 3 Elevation Mat / Sheet 2 Sections	100% Preserved
15M	STR	7890	4	4	400W	Sheet 3 Elevation Mat / Sheet 2 Sections	100% Preserved
15M	STR	8790	2	2	400W	Sheet 3 Elevation Mat / Sheet 2 Sections	100% Preserved
15M	STR	9270	2	2	400W	Sheet 3 Elevation Mat / Sheet 2 Sections	100% Preserved

Table 5: Architectural Finishes, Mould Tags & Structural General Notes (W10-45)

Verification of architectural finishes, formwork edge datums, and structural general notes that were omitted or clipped by the macro generator and established by the detailing engineer in Rev 2.

Feature / Item	Macro State (W10-45)	Ground Truth State (Rev 2)	Defect Identified in Macro	Applied Engineering Logic
Architectural Reveals (V-Grooves)	Cutouts and shallow grooves duplicated under generic label	Dedicated dimension line: REVEAL (B): 241 943 931 975 884 3548 497 975	Feature Confusion	Segregated cosmetic surface millwork from structural wall forming to eliminate false

Feature / Item	Macro State (W10-45)	Ground Truth State (Rev 2)	Defect Identified in Macro	Applied Engineering Logic
	'PROFILE RECESS'	373 (Sum = 9,367 mm $\approx$ 9,366 mm)		bucking.
Architectural Paint Specifications	Completely missing; no paint codes, demarcation lines, or finish boundaries	4 distinct paint shades scheduled: Thunder, Silhouette, Etruscan, Simply White + explicit 'NO PAINT' strip	Finishing Data Omission	Mapped exact masking boundaries directly on elevation face for turnkey paint shop execution.
Formwork Edge Tags (Carpentry)	Interior window cutout perimeters and external panel sides completely unmarked	All mould faces standardized: External: END 1 through END 5; Internal: SIDE A through SIDE D	Panel Orientation Ambiguity	Assigned standardized Cartesian identifiers ensuring precise formwork bed setup.
General Notes & Structural Block	Text clipped; structural design values left unpopulated (28 DAY MPa and STRIPPING N blank)	Fully populated criteria: 28-Day Strength 6,000 PSI; Stripping Strength 3,500 PSI; Panel Weight 24,827 LBS; Min Cover 25 mm	Safety Compliance Violation	Populated mandatory engineering release criteria directly from design calculation report.

Table 6: Drawing Metadata & Viewport Transformation Matrix (Macro vs. Rev 2)

Exact geometric sheet parameters and the true affine transformation matrix between the raw macro and Rev 2 for Panel W10-45.

Parameter / Zone	Macro Combined (Rev 0)	Engineer Rev 2 (Ground Truth)	Engineering Transformation Formulation
Sheet Dimensions	1296.00 pt x 864.00 pt (18" x 12")	1224.00 pt x 792.00 pt (17" x 11" ANSI B)	Page border standardized to Tabloid precast drawing format
Total Sheet Allocation	Sheet 1: Congested Elevation Sheet 2: Congested Rebar Sheet 3: Congested BBS Table	Sheet 1: Clean Elevation & Tape Sheet 2: Architectural Detail Cuts Sheet 3: Rebar Mat & Sections Sheet 4: Dedicated BBS Schedule	Expanded from 3 congested sheets to 4 clean shop-ready sheets
Drawing Elevation Scale	Scale 1:50 (0.056693 pt/mm)	Scale 1:75 (0.037795 pt/mm)	Rescaled by factor $s = 50 / 75 = 0.6667$ to fit ANSI B sheet
Panel Point Span (9366 mm)	530.99 pt on Macro Canvas	353.99 pt on Rev 2 Canvas	Shrinkage ratio: $353.99 / 530.99 = 0.6667$ (Exact Affine Match)
Hardware BOM Table Pos	X = 1058.23 pt, Y = 358.99 pt	X = 1023.08 pt, Y = 322.99 pt	Rigid Shift: $dx = -35.15$ pt, $dy = -36.00$ pt (Refutes $dx \approx -378$ pt)
BBS Schedule Pos	Sheet 3: (77.46, 110.00 pt)	Sheet 4: (42.31, 71.96 pt)	Table Shift: $dx = -35.15$ pt, Row compaction = 4.29 pt/row

PART II: Algorithm Repair Specification for Correct Drawing Generation

To enable your software and AI algorithms to generate correct precast shop drawings and perform automated QA without hallucinating phantom shifts or failing coordinate alignments, the following 4 architectural fixes must be deployed.

Fix 1: Implement Affine Viewport Transformation (Scale s + Translation) Instead of Pure Translation

Why the previous algorithm failed:

The previous algorithm assumed all drawing changes were rigid pure translations: $[X', Y'] = [X + \Delta X, Y + \Delta Y]$. When the detailing engineer rescaled the panel from 1:50 to 1:75, every coordinate shrank by 0.6667x relative to the viewport center. A pure translation model erroneously interpreted this 33.3% geometric contraction as massive coordinate drift.

The Correct Mathematical Formula:

Any point $P = (X, Y)$ in Macro is mapped to $P' = (X', Y')$ in Rev 2 via a 2D Affine Transformation Matrix:

$$\begin{bmatrix} X' \\ Y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & t_x \\ 0 & s_y & t_y \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}$$

Where $s_x = s_y = S_{rev2} / S_{macro} = (1/75) / (1/50) = 50 / 75 = 0.6667$.

Fix 2: Enforce Zone-Constrained Semantic Masking (Preventing Table vs. Callout Cross-Matching)

Why the previous algorithm failed:

The unconstrained script searched for hardware mark strings across the entire sheet without restricting search regions. It found elevation callouts (e.g. P-205 at $X = 632.3$ pt) before finding table cells at $X = 1023.08$ pt, and falsely reported that the BOM table shifted by $dx \approx -378.5$ pt.

The Required Code Repair:

Partition each drawing page into 4 mutually exclusive spatial zones before matching:

1. SCHEDULE_TABLE: Bound $[X0 \geq 1000, 320 \leq Y0 \leq 550]$ (Sheet 1 BOM) or $[X0 \leq 600, Y0 \leq 300]$ (Sheet 4 BBS)
2. TITLE_BLOCK: Bound $[X0 \geq 1000, Y0 \geq 590]$
3. GENERAL_NOTES: Bound $[X0 \geq 1000, Y0 < 320]$
4. STRUCTURAL_CANVAS: Bound $[X0 < 1000]$ (All panel geometry, rebar, dimensions, and callouts)

Rule: Elements in SCHEDULE_TABLE must ONLY match against SCHEDULE_TABLE elements. Cross-zone matches are strictly forbidden.

Fix 3: Upgrade Token Matching to Word-Boundary Regex (Eliminating Substring Hallucinations)

Why the previous algorithm failed:

The script searched for panel thickness '200' using generic substring containment: if '200' in text. It matched '12000LB FORGED ANCHOR' because '12000' contains '200'. It then reported that panel thickness moved by $(-35.15, -36.00)$.

The Required Code Repair:

Use exact whole-word regular expression tokenizers:

```
pattern = re.compile(rf'\b{re.escape(token)}\b')
```

This guarantees that anchor capacities (12000LB), rebar lengths (3200 mm), and dimension values (200 mm) remain strictly isolated.

Fix 4: Two-Pass Automated Drawing Generation Pipeline

To enable your automated drawing generation software (Tekla Open API / AI Agent) to create human-grade shop drawings:

- Pass 1: Primary Elevation Layout (Sheet 1)
 - Place panel outer boundary and window cutouts at scale 1:75.
 - Enforce the 7-Tier Dimension Hierarchy (Lifters -> Splicers -> Bracing -> Hardware -> BOM -> Openings -> Overall 9366 mm).
 - Solve 1D interstitial spans to unify broken window strings into single continuous chains (2115 | 975 | 4928 | 975 | 373 mm).

- Stagger hardware leaders neatly outward to the right margin ($X \approx 632.3$ pt).
- Pass 2: Section Cuts & Rebar Extraction (Sheets 2, 3 & 4)
 - Route all architectural reveals, drip edges, and sill profiles to Sheet 2.
 - Migrate rebar mat layout and diagonal opening trimmers to dedicated Sheet 3.
 - Automatically compile the Bar Bending Schedule (BBS) into dedicated Sheet 4 with row shift locked to $dx = -35.15$ pt.

FINAL TAKEAWAY & ML TRAINING CERTIFICATION

SUMMARY FOR YOUR DEVELOPMENT TEAM:

1. Precast Panel W10-45 geometry and Rev 2 drawings are 100% physically correct.
2. W10-45 is the exact mirrored (opposite hand) counterpart to W10-175, with opening sequence: 2115 | 975 | 4928 | 975 | 373 mm.
3. The Hardware BOM Table shift is exactly $dx = -35.15$ pt, $dy = -36.00$ pt ($\sigma = 0.000$ pt), matching panels W10-67, W10-43, and W10-175.
4. Deploying Affine Scale Modeling ($s = 0.6667$), Zone Masking, and the 7-Tier Hierarchy certifies your automated pipeline for production.